

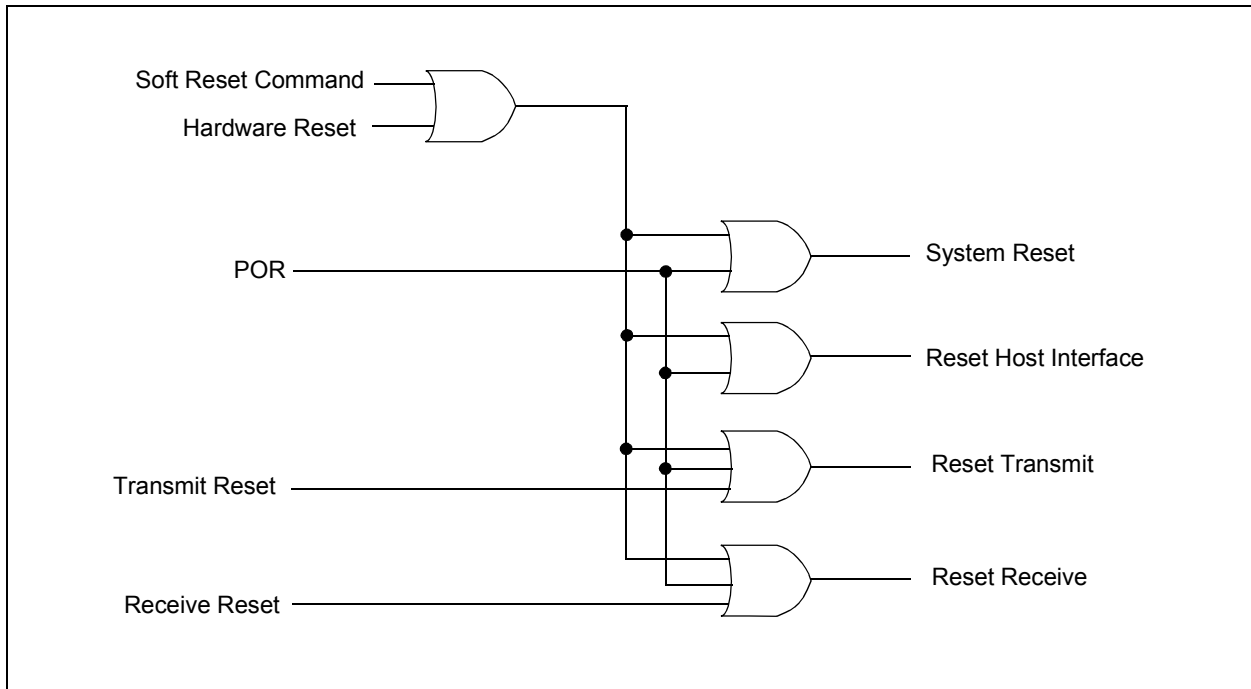
11.0 RESET

The ENC28J60 differentiates between various kinds of Reset:

- Power-on Reset (POR)
- System Reset
- Transmit Only Reset
- Receive Only Reset
- Miscellaneous MAC and PHY Subsystem Resets

A simplified block diagram of the On-Chip Reset Circuit is shown in Figure 11-1.

FIGURE 11-1: ON-CHIP RESET CIRCUIT



11.1 Power-on Reset (POR)

A Power-on Reset pulse is generated on-chip whenever VDD rises above a certain threshold. This allows the device to start in the initialized state when VDD is adequate for operation.

The POR circuitry is always enabled. As a result, most applications do not need to attach any external circuitry to the $\overline{\text{RESET}}$ pin to ensure a proper Reset at power-up. The $\overline{\text{RESET}}$ pin's internal weak pull-up will maintain a logical high level on the pin during normal device operation.

To ensure proper POR operation, a minimum rise rate for VDD is specified (parameter D003). The application circuit must meet this requirement to allow the Oscillator Start-up Timer and CLKOUT functions to reset properly.

After a Power-on Reset, the contents of the dual port buffer memory will be unknown. However, all registers will be loaded with their specified Reset values. Certain portions of the ENC28J60 must not be accessed immediately after a POR. See **Section 2.2 "Oscillator Start-up Timer"** for more information.

11.2 System Reset

The System Reset of ENC28J60 can be accomplished by either the $\overline{\text{RESET}}$ pin, or through the SPI interface.

The $\overline{\text{RESET}}$ pin provides an asynchronous method for triggering an external Reset of the device. A Reset is generated by holding the $\overline{\text{RESET}}$ pin low. The ENC28J60 has a noise filter in the $\overline{\text{RESET}}$ path which detects and ignores small pulses of time, t_{RSTLOW} , or less. When the $\overline{\text{RESET}}$ pin is held high, the ENC28J60 will operate normally.

The ENC28J60 can also be reset via the SPI using the System Reset Command. See **Section 4.0 "Serial Peripheral Interface (SPI)"**.

The $\overline{\text{RESET}}$ pin will not be driven low by any internal Resets, including a System Reset command via the SPI interface.

After a System Reset, all PHY registers should not be read or written to until at least 50 μs have passed since the Reset has ended. All registers will revert to their Reset default values. The dual port buffer memory will maintain state throughout the System Reset.

11.3 Transmit Only Reset

The Transmit Only Reset is performed by writing a '1' to the TXRST bit in the ECON1 register using the SPI interface. If a packet was being transmitted when the TXRST bit was set, the hardware will automatically clear the TXRTS bit and abort the transmission. This action resets the transmit logic only. The System Reset automatically performs the Transmit Only Reset. Other register and control blocks, such as buffer management and host interface, are not affected by a Transmit Only Reset event. When the host controller wishes to return to normal operation, it should clear the TXRST bit.

11.4 Receive Only Reset

The Receive Only Reset is performed by writing a '1' to the RXRST bit in the ECON1 register using the SPI interface. If packet reception was enabled (the RXEN bit was set) when RXRST was set, the hardware will automatically clear the RXEN bit. If a packet was being received, it would be immediately aborted. This action resets receive logic only. The System Reset automatically performs Receive Only Reset. Other register and control blocks, such as the buffer management and host interface blocks, are not affected by a Receive Only Reset event. When the host controller wishes to return to normal operation, it should clear the RXRST bit.

11.5 PHY Subsystem Reset

The PHY module may be reset by writing a '1' to the PRST bit in the PHCON1 register (Register 11-1). All the PHY register contents will revert to their Reset defaults.

Unlike other Resets, the PHY cannot be removed from Reset immediately after setting PRST. The PHY requires a delay, after which the hardware automatically clears the PRST bit. After a Reset is issued, the host controller should poll PRST and wait for it to become clear before using the PHY.

REGISTER 11-1: PHCON1: PHY CONTROL REGISTER 1

R/W-0	R/W-0	U-0	U-0	R/W-0	R/W-0	U-0	R/W-0
PRST	PLOOPBK	—	—	PPWRSV	r	—	PDPXMD ⁽¹⁾
bit 15				bit 8			

R/W-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
r	—	—	—	—	—	—	—
bit 7				bit 0			

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 15 **PRST:** PHY Software Reset bit

1 = PHY is processing a Software Reset (automatically resets to '0' when done)

0 = Normal operation

bit 14 **PLOOPBK:** PHY Loopback bit

1 = All data transmitted will be returned to the MAC. The twisted-pair interface will be disabled.

0 = Normal operation

bit 13-12 **Unimplemented:** Read as '0'

bit 11 **PPWRSV:** PHY Power-Down bit

1 = PHY is shut down

0 = Normal operation

bit 10 **Reserved:** Maintain as '0'

bit 9 **Unimplemented:** Read as '0'

bit 8 **PDPXMD:** PHY Duplex Mode bit⁽¹⁾

1 = PHY operates in Full-Duplex mode

0 = PHY operates in Half-Duplex mode

bit 7 **Reserved:** Maintain as '0'

bit 6-0 **Unimplemented:** Read as '0'

Note 1: Reset values of the Duplex mode/status bits depend on the connection of the LED to the LEDB pin (see **Section 2.6 "LED Configuration"** for additional details).

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NOTES: